

# Root systems

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## 0.1 Root pairings

### 0.1.1 Definitions

Recall that a *perfect pairing* is a quadruple  $(R, M, N, \mathcal{L})$  where  $R$  is a ring,  $M$  and  $N$  are  $R$ -modules, and  $\mathcal{L} : M \times N \rightarrow R$  is a bilinear map such that the induced maps  $M \rightarrow \text{Hom}_R(N, R)$  and  $N \rightarrow \text{Hom}_R(M, R)$  are isomorphisms of  $R$ -modules.

If  $I$  is a set, denote by  $\text{Perm}(I)$  the group of permutations of  $I$ .

**Definition 1.** A *root pairing* is a tuple  $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$  where  $(R, M, N, \mathcal{L})$  is a perfect pairing,  $I$  is a set,  $\alpha : I \rightarrow M, i \mapsto \alpha_i$ ,  $\alpha^\vee : I \rightarrow N, i \mapsto \alpha_i^\vee$ , and  $s : I \rightarrow \text{Perm}(I), i \mapsto s_i$ , with the conditions that

1.  $\alpha, \alpha^\vee$  are injective,
2. For all  $i \in I$ ,  $\mathcal{L}(\alpha_i, \alpha_i^\vee) = 2$ ,
3. For all  $i, j \in I$ , we have

$$\begin{aligned}\alpha_{s_i(j)} &= \alpha_j - \mathcal{L}(\alpha_j, \alpha_i^\vee)\alpha_i, \\ \alpha_{s_i(j)}^\vee &= \alpha_j^\vee - \mathcal{L}(\alpha_i, \alpha_j^\vee)\alpha_i^\vee.\end{aligned}$$

### 0.1.2 Root pairing of type A

Consider the perfect pairing  $(R, M, N, \mathcal{L}) = (\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, (\cdot, \cdot))$  where  $(\cdot, \cdot)$  is the canonical pairing between  $\mathbb{Z}^n$  and its algebraic dual  $(\mathbb{Z}^n)^* := \text{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z})$ , i.e. the dot product.

We denote by  $e_1, \dots, e_n$  the standard basis of  $\mathbb{Z}^n$  and by  $e_1^*, \dots, e_n^*$  the dual basis of  $(\mathbb{Z}^n)^*$ .

Write  $[1, n] = \{1, \dots, n\}$ . An *interval* in  $[1, n]$  is a subset of the form  $[i, j] \stackrel{\text{denotes}}{=} \{i, i+1, \dots, j\}$  for some  $1 \leq i \leq j \leq n$ . We denote by  $I_n$  the set of intervals in  $[1, n]$ . Write  $[i] := [i, i]$  for  $1 \leq i \leq n$ .

A *signed interval* in  $[1, n]$  is an element of  $I_n := I_n \times \{1, -1\}$ . We denote by  $I_n^+$  the set of signed intervals of the form  $(J, 1)$  and by  $I_n^-$  the set of signed intervals of the form  $(J, -1)$ . It is clear that  $I_n = I_n^+ \sqcup I_n^-$ . Throughout the rest of this file, we write  $J$  for the signed interval  $(J, 1)$  and  $-J$  for  $(J, -1)$ .

Let  $A = (a_{i,j})$  be the Cartan matrix of type  $A_n$  defined by  $a_{i,i} = 2$  for all  $i \in [1, n]$ ,  $a_{i,j} = -1$  if  $|i - j| = 1$ , and  $a_{i,j} = 0$  otherwise.

Define

$$\begin{aligned}\alpha : I_n &\rightarrow \mathbb{Z}^n, & \forall J \in I_n^+, \quad \alpha_{\pm J} &= \pm \left( \sum_{j \in J} A e_j \right), \\ \alpha^\vee : I_n &\rightarrow (\mathbb{Z}^n)^*, & \forall J \in I_n^+, \quad \alpha_{\pm J}^\vee &= \pm \left( \sum_{j \in J} e_j^* \right).\end{aligned}$$

Finally, define  $s : I_n^+ \rightarrow \text{Perm}(I_n)$  as follows. Fix  $J, K \in I_n^+$  and set  $L = (J \cup K) \setminus (J \cap K)$ . It is clear that  $J, K, L$  satisfy exactly one of the following four conditions.

**R1.**  $L = \emptyset$ .

**R2.**  $L \in I_n$  and  $K \subset J$ , and

**R3.**  $L \in I_n$  and  $K \not\subset J$ .

**R4.**  $L \notin I_n \cup \{\emptyset\}$ .

Then,

$$s_J(K) = \begin{cases} -K & \text{in Case \textbf{R1},} \\ -L & \text{in Case \textbf{R2},} \\ L & \text{in Case \textbf{R3}.} \\ K & \text{in Case \textbf{R4},} \end{cases} \quad (1)$$

If  $J = [i, j]$  and  $K = [k, l]$ , then (1) can be rewritten as

$$s_{[i,j]}[k, l] = \begin{cases} -[k, l] & \text{if } (i, j) = (k, l) \\ -[l + 1, j] & \text{if } i = k, j > l \\ [j + 1, l] & \text{if } i = k, j < l \\ -[i, k - 1] & \text{if } i < k, j = l \\ [k, i - 1] & \text{if } i > k, j = l \\ [k, j] & \text{if } i = l + 1 \\ [i, l] & \text{if } k = j + 1 \\ [k, l] & \text{otherwise.} \end{cases} \quad (2)$$

We extend  $s_J$  to  $I_n$  by setting  $s_J(-K) = -s_J(K)$  for all  $K \in I_n^+$ . We extend  $s$  to  $I_n$  by setting  $s_{-J} = s_J$  for all  $J \in I_n^+$ .

**Lemma 2.** *For all  $J$ , we have  $s_J \in \text{Perm}(I_n)$ .*

*Proof.* Fix  $J \in I_n$ . We prove that  $s_J^2 = \text{id}$  by a straightforward case-by-case check using the definition of  $s_J$  and the four cases above. Then  $s_J^{-1} = s_J$ , and our claim is proved.  $\square$

**Lemma 3.**  $\alpha : I_n \rightarrow \mathbb{Z}^n$  is injective, i.e.  $\alpha_J = \alpha_K \implies J = K$

*Proof.* Write

$$\begin{aligned} \alpha_J &= \varepsilon_J \left( \sum_{j \in +J} A e_j \right), \varepsilon_J \in \{-1, 1\} \\ \alpha_K &= \varepsilon_K \left( \sum_{k \in +K} A e_k \right), \varepsilon_K \in \{-1, 1\} \end{aligned}$$

Then  $\alpha_J = \alpha_K \implies \varepsilon_J (\sum_{j \in +J} A e_j) - \varepsilon_K (\sum_{k \in +K} A e_k) = 0$ . This reduces to

$$\sum_{k \in +J \cup +K} c_k A e_k = 0$$

with the following cases:

$$c_k = \begin{cases} \varepsilon_J & \text{if } k \in +J, k \notin +K \\ -\varepsilon_K & \text{if } k \notin +J, k \in +K \\ \varepsilon_J - \varepsilon_K & \text{if } k \in +J, k \in +K \\ 0 & \text{if } k \notin +J, k \notin +K \end{cases}$$

**Lemma 3.1.**  $A_n$  is invertible for all  $n \in \mathbb{N}$ , i.e.  $\det(A_n) \neq 0$

*Proof.* Since  $A_n$  is a tridiagonal matrix with diagonal entries = 2 and off-diagonal entries = -1,  $\det(A_n) = 2\det(A_{n-1}) - \det(A_{n-2})$ . In particular,  $\det(A_n) - \det(A_{n-1}) = \det(A_{n-1}) - \det(A_{n-2})$  for all  $n$ .  $\det(A_1) = 2$  and  $\det(A_2) = 3$ , so  $A_n = n + 1$  for all  $n$ , which is clearly  $\neq 0$ .  $\square$

**Corollary 3.1.**  $A_n e_k$  is linearly independent  $\forall k \leq n \in \mathbb{N}$ .

*Proof.* It is known that a set of linearly independent vectors multiplied by an invertible matrix is still linearly independent.  $\square$

By the above lemma,  $Ae_k$  is linearly independent, so  $\forall k \in +J \cup +K, c_k = 0$ . Therefore,  $\forall k \in +J \cup +K, k$  must be in both  $+J$  and  $+K$ . In this case  $\varepsilon_J - \varepsilon_K = 0$ , so  $\varepsilon_J = \varepsilon_K$ , which means  $J = K$ .  $\square$

**Lemma 4.**  $\alpha^\vee$  is injective, i.e.  $\alpha_J^\vee = \alpha_K^\vee \implies J = K$

*Proof.* Clearly, the set of  $e_i^*$  is linearly independent, so the same proof as Lemma 3 holds.  $\square$

**Lemma 5.**  $\forall J \in I_n, (\alpha_J^\vee, \alpha_J) = 2$

*Proof.* Write

$$\alpha_J = \varepsilon_J \left( \sum_{j \in +J} Ae_j \right), \varepsilon_J \in \{-1, 1\}$$

Then

$$\begin{aligned} (\alpha_J^\vee, \alpha_J) &= (\varepsilon_J \left( \sum_{j \in +J} e_j^* \right), \varepsilon_J \left( \sum_{k \in +J} Ae_k \right)) \\ &= \sum_{j \in +J} \sum_{k \in +J} (e_j^*, Ae_k) \\ &= \sum_{j \in +J} \sum_{k \in +J} A_{j,k} \end{aligned}$$

For length  $L = |J|$  of the interval, there are  $L$  diagonal terms in the sum,  $2(L-1)$  pairs of diagonal-adjacent indices, each contributing 2 and  $-1$  respectively to the sum (by the definition of the Cartan matrix  $A$  of type  $A_n$ ). Therefore,  $\sum_{j \in +J} \sum_{k \in +J} A_{j,k} = 2L - 2(L-1) = 2$   $\square$

**Lemma 6.** For all  $J, K \in I_n$ ,  $\alpha_{s_J(K)} = \alpha_K - (\alpha_K^\vee, \alpha_J) \alpha_J$

*Proof.* Let  $J = \text{sign}(J)J$ , where  $J_+ = [i, j]$ ,  $K = \text{sign}(K)K_+$ , where  $K_+ = [k, l]$ . Then,

$$\begin{aligned} (\alpha_K^\vee, \alpha_J) &= \left( \sum_{u=k}^l e_u^*, \sum_{v=i}^j Ae_v \right) \\ &= \sum_{u=k}^l \sum_{v=i}^j (e_u^*, Ae_v) \\ &= \sum_{u=k}^l \sum_{v=i}^j A_{uv} \\ &= 2(\# \text{ of pairs } u = v) - (\# \text{ of pairs } v = u + 1) - (\# \text{ of pairs } v = u - 1) \end{aligned}$$

by the definition of  $A$ . Then there are three cases:

**C1.**  $u = v \implies v \in [i, j] \cap [k, l] \implies (\# \text{ of pairs } u = v) = |[i, j] \cap [k, l]| =: L$

**C2.**  $u = v + 1 \implies v \in [i, j] \cap [k-1, l-1] \implies (\# \text{ of pairs } u = v + 1) = L - \delta_{i,k} - \delta_{i,l+1}$ , and

**C3.**  $u = v - 1 \implies v \in [i, j] \cap [k + 1, l + 1] \implies (\# \text{ of pairs } u = v - 1) = L - \delta_{j,l} + \delta_{j,k-1}.$

Therefore

$$\begin{aligned} (\alpha_K^\vee, \alpha_J) &= \left( \sum_{u=k}^l e_u^*, \sum_{v=i}^j A e_v \right) \\ &= 2L - (L - \delta_{i,k} - \delta_{j,l} + \delta_{i,l+1}) - (L - \delta_{i,k} - \delta_{j,l} + \delta_{j,k-1}) \\ &= \delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{j,k-1} \end{aligned}$$

In particular, the cases for  $(\alpha_K^\vee, \alpha_J)$  are as follows:

$$\begin{cases} 2 & \text{if } i = k, j = l \\ 1 & \text{if } i = k, j = l - 1 \text{ or } i = k + 1, j = l \\ -1 & \text{if } i = l + 1, k = j + 1 \\ 0 & \text{otherwise} \end{cases}$$

Applying these cases to the formula  $\alpha_K - (\alpha_K^\vee, \alpha_J) \alpha_J$  (as well as the cases of  $\text{sign}(J)$  and  $\text{sign}(K)$ ), it can be easily verified (case-by-case) that they are equivalent to  $\alpha_{s_J(K)}$  using the definition (2).  $\square$

**Lemma 7.** For all  $J, K \in I_n$ ,  $\alpha_{s_J(K)}^\vee = \alpha_K^\vee - (\alpha_J^\vee, \alpha_K) \alpha_J^\vee$

*Proof.*

$\square$

**Theorem 8.** The 8-tuple  $(\mathbb{Z}, \mathbb{Z}^l, (\mathbb{Z}^l)^*, (\cdot, \cdot), I_n, \alpha, \alpha^\vee, s)$  is a root pairing.

*Proof.*

$\square$